Beyond Deterministic Selection: A Probabilistic Approach for Feature Subset Selection in Medical Data

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***Abstract*—Feature selection is a crucial step in traditional machine learning pipelines, especially for high dimensional datasets where irrelevant or redundant features can degrade model performance. Unlike deterministic feature selection techniques, we propose a novel approach to assign probabilities to features based on information theoretic metrics such as mutual information, interaction scores, linear correlation, information gain, Spearman rank correlation, and Shannon entropy. This gives each feature a chance of being chosen based on how important it seems. After sampling features, multiple ML classifiers are trained on these subsets of medical data. To see how well our method works, we ran experiments on five medical dataset. We compared its performance against using all features and random feature selection. The results showed that our method not only performed better but also we see that we can achieve great results with less number of features if sampled using information metric . This reduction led to faster training times and lower storage requirements, making our method a more efficient way to build accurate models. Furthermore, by analyzing the distribution of selected features across the ensembles, we gain valuable insights into the stability and importance of individual features.**

I. Introduction

Feature selection is a critical step in machine learning pipelines, particularly for high-dimensional datasets where irrelevant or redundant features can hinder model performance [1]. Traditional methods often select a fixed subset of features, which may not be optimal in scenarios where feature importance can vary. This paper introduces a novel probabilistic framework for feature subset selection that addresses these limitations.

Our approach leverages information-theoretic metrics such as mutual information, interaction scores, linear correlation, information gain, Spearman rank correlation, and Shannon entropy to assign probabilities to features. This probabilistic assignment allows for flexible feature subset sampling, enabling the exploration of diverse feature combinations and improving the robustness of the feature selection process.

We integrate this probabilistic framework with ensemble learning methods, training multiple models on different feature subsets sampled according to their assigned probabilities. This ensemble approach not only enhances model performance but also provides valuable insights into the stability and importance of individual features by analyzing the distribution of selected features across the ensemble.

Experimental results on multiple medical datasets demonstrate that our method outperforms both using all features and random feature selection. Furthermore, we observe significant improvements in model performance with a reduced number of features, leading to faster training times and lower storage requirements. This probabilistic framework offers a more efficient and effective approach to feature selection, enhancing model performance and interpretability across various machine learning tasks.

II. Research Question

In this study we try to understand how do our approach of using various information theoretic metrics for probabilistic feature subset selection impact the performance, generalization, and interpretability of various classifiers, compared to using all the features or random sampling, on high-dimensional datasets.

III. Literature Review

Peng et al. [2] introduced a feature selection method based on mutual information, proposing criteria such as max-dependency, max-relevance, and min-redundancy. While their approach effectively evaluates the relationship between features and target variables, it falls short in capturing interactions between features. Hall [3] developed the Correlation-Based Feature Selection (CFS) algorithm, which selects features by balancing their correlation with the target variable and minimizing inter-correlation among features. This algorithm became a foundational method in filter-based feature selection by addressing redundancy while ensuring relevance. Chandrashekar and Sahin [4] provided a detailed survey categorizing feature selection methods into filter, wrapper, and embedded techniques, highlighting the trade-offs between computational efficiency and the ability to capture feature dependencies. Zou and Hastie [5] contributed the Elastic Net, an embedded feature selection method that integrates

L1 and L2 penalties, offering a balance between sparsity and grouping effects, particularly useful for high-dimensional datasets. In the context of ensemble learning, Ho [6] proposed the Random Subspace Method, which trains classifiers on randomly selected feature subsets to enhance diversity and robustness, paving the way for feature bagging techniques. Saeys et al. [7] extended this discussion with a review of feature selection methods for bioinformatics, emphasizing the challenges of high-dimensional datasets and the advantages of ensemble-based approaches like feature bagging. Shannon [8] laid the theoretical foundation for measuring uncertainty and information content through entropy, a concept widely applied in information-theoretic feature selection methods. Finally, Dietterich [9] explored ensemble learning methods, demonstrating how combining predictions from multiple models can improve robustness and generalization, supporting the idea of integrating feature selection with ensemble techniques. Collectively, these studies highlight the evolution of feature selection techniques and underscore the potential of probabilistic and ensemble-based methods for improving model robustness and efficiency.

IV. Dataset Description

This study used five diverse datasets: the Breast Cancer Dataset (BC), originating from the University Medical Center in Ljubljana, Yugoslavia, and frequently used in machine learning research [10], and the Breast Cancer Wisconsin (BCW) dataset [11]; the Diabetes dataset, that was obtained from two sources: an automatic electronic recording device and paper records [12]; the Heart Failure Dataset (HF), containing medical records of 299 patients with heart failure, encompassing 13 clinical features [13]; and the Liver Disorders Dataset (LD), donated by BUPA Medical Research Ltd. [14]. The general overview can be seen in the Table I. Further, the datasets were pre-processed by removing samples that contained missing feature values or NaN entries.

TABLE I
UCI Datasets Overview

Name	Feature Type	# Instances	# Features
BC	Categorical	286	9
BCW	Numerical	569	30
Diabetes	Categorical, Numerical	1172	20
HF	Numerical	299	12
LD	Categorical, Numerical	345	5

V. Proposed Method

The proposed methodology involves a systematic approach to selecting and evaluating features, followed by optimizing classifier performance. Initially, features are scored using multiple evaluation metrics to approximate their importance. A probabilistic feature selection process is employed to select subsets of features, varying the number of selected features (n) to identify the optimal subset. For each feature subset, classifiers are trained and evaluated using k-fold cross-validation, and a grid search is performed to identify the best hyperparameters for each model. This comprehensive process ensures robust feature selection and model optimization tailored to the characteristics of each dataset.

A. Feature Evaluation Metrics (Scoring methods)

To approximate the probabilities of features in our datasets, several techniques were employed to quantify relationships between variables. These metrics were normalized and interpreted as "probabilities," enabling the selection of the most relevant features based on their influence on target variables.

1) Mutual Information: Mutual information (MI) quantifies the reduction in uncertainty about one variable given knowledge of another. It measures the degree of dependency between the feature and the target variable. In our approach, the MI values were computed and normalized to sum to 1, representing the probabilities of feature importance.

$$I(X;Y) = \sum_{x \in X} \sum_{y \in Y} P(x,y) \log \frac{P(x,y)}{P(x)P(y)}. \tag{1}$$

where:

- $P(x,y)$ is the joint probability of x and y,
- $P(x)$ and $P(y)$ are the marginal probabilities of x and y, respectively.

2) Interaction Scores: Interaction scores were calculated using a Random Forest model. Feature importances were computed as a measure of how much each feature contributes to decision-making. The pairwise interaction scores between features were then calculated by aggregating feature importance values across all trees and normalizing them. The final scores were interpreted as probabilities, summing to 1.

3) Linear Correlation: Linear correlation between features and target variables was measured using the Pearson correlation coefficient. This coefficient quantifies the strength and direction of a linear relationship.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}. \tag{2}$$

where:

- x_i, y_i are the values of the variables in the sample,
- $\bar{x}, \bar{y}$ are the means of the variables.

4) Information Gain: Information gain (IG) measures the reduction in entropy of the target variable Y after observing a feature X. It evaluates the usefulness of a feature for predicting the target.

$$IG(Y|X) = H(Y) - H(Y|X). \tag{3}$$

where:

- $H(Y)$ is the entropy of Y,
- $H(Y|X)$ is the conditional entropy of Y given X.

5) Spearman Rank Correlation: The Spearman rank-order correlation coefficient measures the monotonic relationship between two variables. It is defined as:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}. \tag{4}$$

where:

- d_i is the difference between the ranks of each observation,
- n is the number of observations.

6) Shannon Entropy: Shannon entropy quantifies the uncertainty or randomness in a feature. It measures the expected information content in a random variable.

$$H(X) = -\sum_i P(x_i) \log P(x_i). \tag{5}$$

where $P(x_i)$ is the probability of the i-th outcome.

B. Probabilistic Feature Selection

After computing the feature probabilities using the metrics described above, feature subsets were sampled probabilistically. The goal was to ensure that features with higher importance scores were more likely to be selected, while still allowing some diversity in the subsets.

The probabilistic feature selection process is outlined in Algorithm 1.

Algorithm 1 Probabilistic Feature Selection

- X: Feature matrix of size $n \times m$ (where n is the number of samples and m is the number of features),
- y: Target variable vector of size n,
- k: Number of features to select,
- S: Feature scoring method (e.g., Mutual Information, Linear Correlation, etc.).
- F_k: Subset of k selected features.

Step 1: Compute Feature Scores
Apply the scoring method S to evaluate the importance of each feature with respect to the target variable y. Let s_i denote the score for feature i, where $i \in \{1, 2, \ldots, m\}$.

$$S = \{s_1, s_2, \ldots, s_m\}. \tag{6}$$

Step 2: Normalize Scores to Probabilities
Normalize the scores such that they sum to 1 to interpret them as probabilities:

$$P_i = \frac{s_i}{\sum_{j=1}^{m} s_j} \quad \text{for } i \in \{1, 2, \ldots, m\}. \tag{7}$$

Step 3: Probabilistic Sampling of Features
Initialize an empty set of selected features: $F_k = \{\}$.
$t \leftarrow 1$ to k Sample a feature index f_t from the probability distribution $P = \{P_1, P_2, \ldots, P_m\}$ using random weighted sampling without replacement.
Add the selected feature f_t to F_k:
$F_k = F_k \cup \{f_t\}$.
Step 4: Return Selected Features
Output the set of selected features F_k.

After feature selection, hyperparameter optimization was conducted to fine-tune the classifiers. Grid search strategy was used to explore various hyperparameter configurations systematically. For each classifier, a range of values for key parameters was defined. For Random Forest, we tuned the number of estimators, maximum depth, and minimum samples split. For SVM, we tuned the kernel type, regularization parameter (C), and gamma values. This approach ensured that the classifiers were optimized for each dataset and feature subset, leading to improved predictive performance. Now we look at the experimental analysis of our method on various benchmarking medical dataset from UCI repository.

VI. Results and Discussion

In this section we discuss and compare the results obtained by different classifiers on different dataset using different feature selection methods. Table II shows the hyper parameters used for each classifier during grid search.

TABLE II
GridSearch Hyperparameters

Classifier	Hyperparameter	Possible Values
Random Forest	$n_estimators$	50, 100
	max_depth	10, 20, None
Gradient Boosting	$n_estimators$	50, 100
	$learning_rate$	0.01, 0.1, 1.0
AdaBoost	$n_estimators$	50, 100
	$learning_rate$	0.5, 1.0, 2.0
Logistic Regression	C	0.1, 1.0, 10.0
	$penalty$	l2
SVM	C	1, 5, 10, 20, 100
	$gamma$	scale, auto
	$kernel$	rbf, linear, poly, sigmoid
LDA	$solver$	lsqr, eigen
	$shrinkage$	None, auto

In table III , IV and V we have reported the best results obtained by each classifier using Probabilistic feature selection, all features (here all features means that we train classifier on the original dataset without any feature selection) and random sampling on five different datasets. We perform the experiments using 10 fold cross validation for 10 runs. To have a look at all the results we have provided a drive link which contains all the results for each dataset and different methods by varying the parameter (number of feature to select). You can access the shared folder using Google Drive link.

Our probabilistic models consistently outperform random sampling models across all five datasets or, at the very least, achieve comparable performance. Notably, on the Heart Failure and Liver Disorder datasets, our approach yields significantly higher accuracies compared to both random sampling and using all features, while simultaneously utilizing a much smaller subset of features. These results highlight the effectiveness of our probabilistic technique, which not only improves model accuracy but also reduces computational complexity by simplifying the feature set.

A. Ablation Study

For ablation we study the influence of varying the number of features (n) parameter on classification performance, we

TABLE III
BEST OBTAINED RESULTS (PROBABILISTIC FEATURE SELECTION)

Dataset	Classifier	Best Method	n_features	Accuracy	Std Dev
Breast Cancer	**Random Forest**	**Linear Correlation**	**8**	**0.9649**	**0.0053**
	Gradient Boosting	Linear Correlation	6	0.9571	0.0095
	AdaBoost	Information Gain	8	0.9585	0.0105
	Logistic Regression	Mutual Information	9	0.9561	0.0000
	SVM	Mutual Information	7	0.9595	0.0062
	LDA	Mutual Information	9	0.9463	0.0000
Breast Cancer Wisconsin	**Random Forest**	**Mutual Information**	**8**	**0.9649**	**0.0057**
	Gradient Boosting	Mutual Information	9	0.9561	0.0000
	AdaBoost	Information Gain	8	0.9610	0.0090
	Logistic Regression	Mutual Information	9	0.9561	0.0000
	SVM	Information Gain	7	0.9620	0.0053
	LDA	Shannon Entropy	8	0.9468	0.0055
Diabetic Dataset	**Logistic Regression**	**Shannon Entropy**	**16**	**0.7552**	**0.0056**
	Random Forest	Information Gain	13	0.7009	0.0120
	Gradient Boosting	Information Gain	13	0.7124	0.0102
	AdaBoost	Linear Correlation	19	0.6879	0.0000
	SVM	Information Gain	9	0.6717	0.0037
	LDA	Information Gain	17	0.7399	0.0034
Heart Failure	**Logistic Regression**	**Linear Correlation**	**12**	**0.8167**	**0.0075**
	Random Forest	Linear Correlation	9	0.7856	0.0132
	Gradient Boosting	Linear Correlation	12	0.7556	0.0000
	AdaBoost	Information Gain	6	0.8000	0.0000
	SVM	Information Gain	6	0.7444	0.0000
	LDA	Linear Correlation	12	0.8111	0.0000
Liver Disorder	**AdaBoost**	**Information Gain**	**11**	**0.8593**	**0.0148**
	Random Forest	Mutual Information	12	0.8383	0.0116
	Gradient Boosting	Linear Correlation	13	0.8160	0.0037
	Logistic Regression	Information Gain	7	0.8358	0.0247
	SVM	Linear Correlation	6	0.7000	0.1027
	LDA	Information Gain	7	0.8469	0.0288

TABLE IV
BEST OBTAINED RESULTS (ALL FEATURES)

Dataset	Classifier	n_features	Accuracy	Std Dev
Breast Cancer	Random Forest	9	0.9512	0.0000
	Gradient Boosting	**9**	**0.9561**	**2.22e-16**
	AdaBoost	9	0.9463	2.22e-16
	Logistic Regression	**9**	**0.9561**	**2.22e-16**
	SVM	9	0.9512	0.0000
	LDA	9	0.9463	2.22e-16
Breast Cancer Wisconsin	Random Forest	9	0.9512	0.0000
	Gradient Boosting	**9**	**0.9561**	**2.22e-16**
	AdaBoost	9	0.9463	2.22e-16
	Logistic Regression	**9**	**0.9561**	**2.22e-16**
	SVM	9	0.9512	0.0000
	LDA	9	0.9463	2.22e-16
Diabetic Dataset	Random Forest	19	0.6879	0.0000
	Gradient Boosting	19	0.7052	0.0000
	AdaBoost	19	0.6879	0.0000
	Logistic Regression	**19**	**0.7514**	**0.0000**
	SVM	19	0.6647	0.0000
	LDA	19	0.7197	0.0000
Heart Failure	Random Forest	12	0.7667	0.0000
	Gradient Boosting	12	0.7000	0.0000
	AdaBoost	12	0.7222	0.0000
	Logistic Regression	12	0.8000	0.0000
	SVM	12	0.5889	1.11e-16
	LDA	**12**	**0.8111**	**0.0000**
Liver Disorder	Random Forest	13	0.7654	0.0000
	Gradient Boosting	13	0.7901	0.0000
	AdaBoost	13	0.8148	1.11e-16
	Logistic Regression	13	0.8272	1.11e-16
	SVM	13	0.6420	0.0000
	LDA	**13**	**0.8395**	**0.0000**

TABLE V
BEST OBTAINED RESULTS (RANDOM SAMPLING)

Dataset	Classifier	n_features	Accuracy	p-value
Breast Cancer	**Random Forest**	**7**	**0.9624**	**0.0058**
	Gradient Boosting	9	0.9561	0.0000
	AdaBoost	8	0.9517	0.0086
	Logistic Regression	9	0.9561	0.0000
	SVM	7	0.9595	0.0062
	LDA	6	0.9537	0.0066
Breast Cancer Wisconsin	**Random Forest**	**8**	**0.9615**	**0.0077**
	Gradient Boosting	9	0.9561	0.0000
	AdaBoost	9	0.9463	0.0000
	Logistic Regression	9	0.9561	0.0000
	SVM	7	0.9566	0.0051
	LDA	9	0.9463	0.0000
Diabetic Dataset	Random Forest	19	0.6864	0.0067
	Gradient Boosting	19	0.7014	0.0023
	AdaBoost	19	0.6879	0.0000
	Logistic Regression	**19**	**0.7497**	**0.0019**
	SVM	19	0.6647	0.0000
	LDA	19	0.7197	0.0000
Heart Failure	Random Forest	12	0.7656	0.0105
	Gradient Boosting	12	0.7000	0.0000
	AdaBoost	12	0.7222	0.0000
	Logistic Regression	12	0.7911	0.0044
	SVM	12	0.5889	0.0000
	LDA	**12**	**0.8111**	**0.0000**
Liver Disorder	Random Forest	12	0.8148	0.0207
	Gradient Boosting	13	0.7901	0.0055
	AdaBoost	12	0.8185	0.0000
	Logistic Regression	12	0.8333	0.0138
	SVM	13	0.6420	0.0000
	LDA	**13**	**0.8395**	**0.0000**

varied n and recorded the mean accuracy and standard deviation across multiple runs. The results were compared for different classifiers and feature evaluation methods.

Across the datasets, we observe distinct trends as the number of features increases. For the Heart Failure dataset (Figure 1), Logistic Regression achieved the highest accuracy (**0.8167**) using the Linear Correlation method with 12 features. Accuracy for Logistic Regression improved with increasing features up to the optimal range, after which additional features resulted in stagnation. Random Forest followed with an accuracy of **0.7856** using 9 features with the same method. SVM, in contrast, showed weaker performance with **0.7444** accuracy using 6 features and the Information Gain method, highlighting its sensitivity to feature count and selection technique. Notably, AdaBoost exhibited robustness in performance, maintaining stable accuracy across varying feature counts. For the Liver Disorder dataset (Figure 1), AdaBoost delivered the highest accuracy (**0.8593**) using Information Gain with 11 features, while Random Forest achieved **0.8383** accuracy with Mutual Information and 12 features. Increasing the number of features beyond these optimal thresholds resulted in either minimal gains or slight performance degradation, as irrelevant or redundant features likely reduced model generalization. SVM, with **0.7000** accuracy using Linear Correlation and 6 features, exhibited significant variability, reinforcing that its performance is highly contingent on proper feature selection and optimal feature count.

Across all datasets, feature selection methods like Mutual Information and Linear Correlation consistently enabled higher accuracies, suggesting their effectiveness in isolating impactful features. Conversely, Random Sampling and Shannon Entropy underperformed, indicating their limited ability to capture feature relevance. These findings underscore the importance of carefully balancing the feature count: while too few features may omit critical information, an excess can introduce noise, underscoring the necessity of evaluating performance over a range of feature counts to identify the optimal subset.

VII. T-TEST ANALYSIS

The paired t-tests conducted across datasets provided key insights into the performance differences between feature selection methods for various classifiers. Statistically significant results ($p < 0.05$) highlighted the superiority of certain methods. For instance, in the **Heart Failure** dataset, *Linear Correlation* significantly outperformed *Information Gain* for the Random Forest classifier ($t = 8.70$, $p = 0.0001$, $n = 9$), indicating that the inclusion of fewer, highly correlated features improved model accuracy. Similarly, in the **Liver Disorder** dataset, *Information Gain* was notably better than *Shannon Entropy* for AdaBoost ($t = 3.99$, $p = 0.005$, $n = 11$), reflecting its ability to prioritize the most informative features.

In contrast, several comparisons yielded non-significant results ($p > 0.05$). For instance, in the **Breast Cancer** dataset, comparisons between *Mutual Information* and *Shannon Entropy* for Logistic Regression showed minimal differences in

accuracy ($t = -0.25$, $p = 0.807$, $n = 9$). Similarly, in the **Diabetic Dataset**, SVM showed no significant improvement when using *Mutual Information* over *Linear Correlation* ($t = 0.14$, $p = 0.889$, $n = 9$). These non-significant results often indicate either the inherent robustness of certain classifiers to feature selection methods or limited variability in feature importance across methods, leading to similar performance outcomes. For instance, tree-based classifiers like Random Forest tend to be less sensitive to feature selection methods compared to linear models, which rely more heavily on feature prioritization.

The t-test analysis provides strong evidence that feature selection significantly influences model performance, particularly for datasets with complex decision boundaries. Methods such as *Information Gain* and *Mutual Information* consistently demonstrated their effectiveness across most classifiers, especially for datasets with higher feature variability. However, non-significant results underscore the importance of context; in datasets with highly homogeneous or less discriminative features, simpler methods like *Linear Correlation* can perform comparably to more sophisticated ones. This highlights the need to carefully balance computational complexity and expected performance gains when selecting feature selection methods. Future work could explore hybrid approaches that combine the strengths of multiple methods to address such scenarios.

Table VI presents a selected subset of the tests we conducted, showcasing both significant and non-significant results. We intentionally limited the scope of the table to maintain clarity and readability. As for the t-tests, they were conducted by evaluating our models over 10 runs over 10 fold cross-validation to ensure robust and reliable results.

A. Frequency analysis

The frequency of selected features across datasets highlights the importance of certain attributes in classification tasks. For the Breast Cancer dataset, **Bare Nuclei** was the most frequently selected feature, followed by **Normal Nucleoli**, indicating their strong predictive capability. Similarly, in the Liver Disorder dataset, **Chest Pain** and **Oldpeak** were dominant, emphasizing their relevance in identifying liver related anomalies. For the Heart Failure dataset, **Ejection Fraction** and **Time** were consistently prioritized, reflecting their critical role in predicting heart conditions. We have plotted some plots in Figure 2

These results are indicative that the probabilistic selection method's ability to prioritize meaningful features while filtering out less impactful ones, which enhances model interpretability and performance. But at the same time proper domain knowledge required to assess this section further.

VIII. Conclusion

In this work we presented a probabilistic feature selection method that leverages multiple information-theoretic metrics—such as mutual information, linear correlation, and information gain—to assign selection probabilities to features. Across five medical datasets, our approach outperformed or equalled the performance of using all features or random sampling, often with fewer features. By systematically sampling multiple subsets, it also provided deeper insights into each feature's relative importance and model stability. Paired t-tests confirmed statistically significant improvements in most scenarios, highlighting benefits in terms of accuracy, generalization, and interpretability.

Future work includes incorporating more advanced metrics, evaluating the approach on larger and more diverse clinical datasets, and extending it to deep learning architectures. Such expansions could further boost the scalability and robustness of probabilistic feature selection, ultimately enhancing machine learning pipelines for medical applications that demand efficient, accurate, and interpretable models.

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Author Contributions

All team members contributed equally to the report writing and the development of the research methodology. The detailed contribution can be seen in the Table VIII.

Use of AI Tools

During the preparation of this work, the authors used generative AI tools to review the grammatical correctness of sentences and ensure adherence to the academic style. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the work.

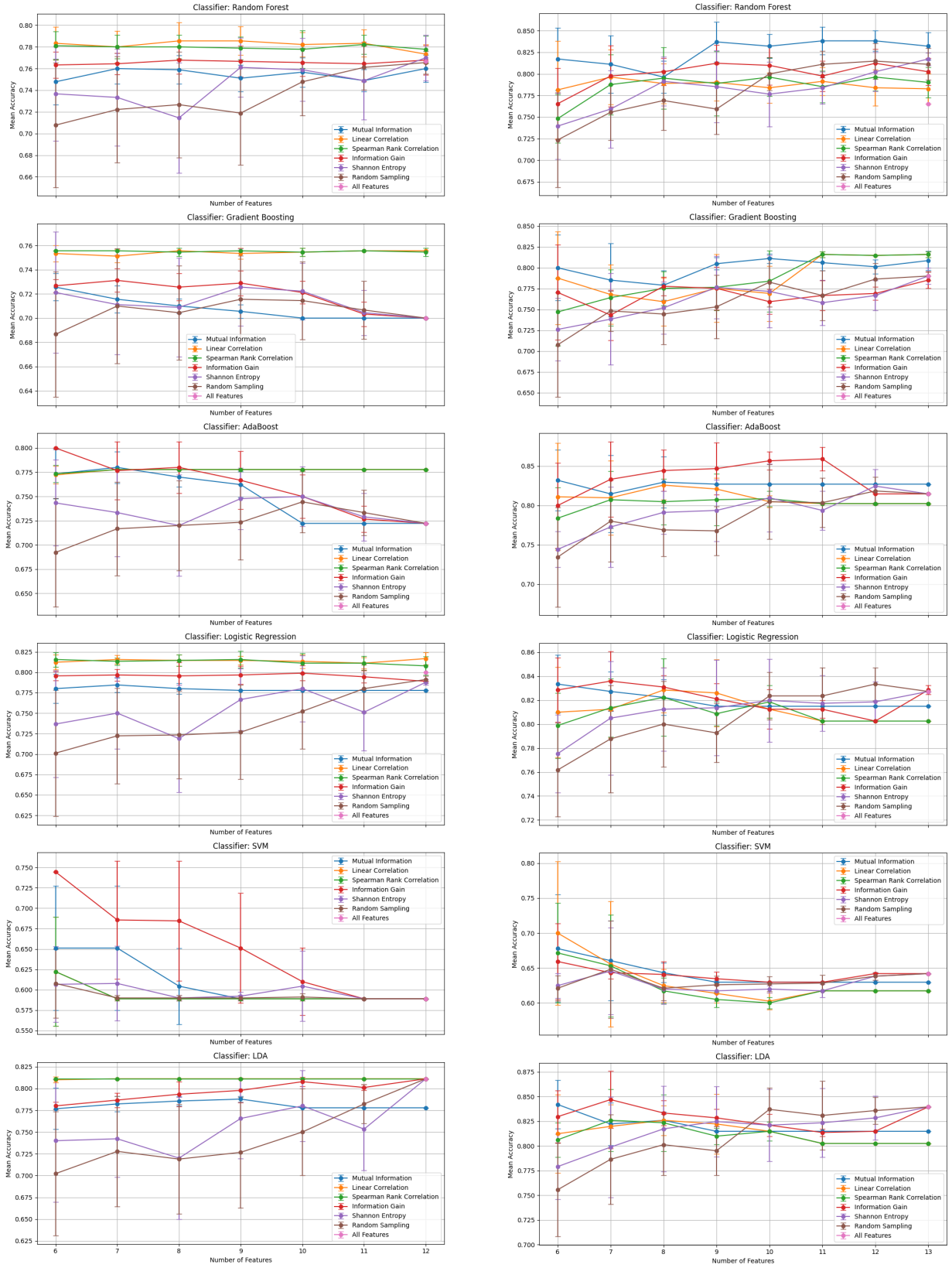


Fig. 1. Accuracy of ensemble models for feature subsets of varying sizes, selected using the specified relationship measurement techniques (left - Heart Failure, right - Leaver Disorder).

TABLE VI
SUMMARY OF SIGNIFICANT AND NON-SIGNIFICANT T-TEST RESULTS

Dataset	**Classifier**	**Method Comparison**	**t-stat**	**p-value**	**n**	**Significance**
Heart Failure	Random Forest	Linear Correlation vs Information Gain	**8.70**	**0.0001**	9	Significant
Liver Disorder	AdaBoost	Information Gain vs Shannon Entropy	**3.99**	**0.005**	11	Significant
Breast Cancer	Logistic Regression	Mutual Information vs Shannon Entropy	**-3.12**	**0.026**	9	Significant
Diabetic Dataset	SVM	Mutual Information vs Linear Correlation	-0.14	0.889	9	Non-Significant
Breast Cancer Wisconsin	Gradient Boosting	Linear Correlation vs Shannon Entropy	1.25	0.234	9	Non-Significant
Liver Disorder	Random Forest	Linear Correlation vs Shannon Entropy	1.95	0.091	12	Non-Significant

TABLE VII
TOP METHODS AND THEIR PERFORMANCE ACROSS DATASETS

Dataset	**Classifier**	**Top Method**	**n**	**Accuracy**	**Std Dev**
Breast Cancer	Random Forest	Linear Correlation	8	**0.9649**	0.0053
Heart Failure	Logistic Regression	Linear Correlation	12	**0.8167**	0.0075
Liver Disorder	AdaBoost	Information Gain	11	**0.8593**	0.0148
Diabetic Dataset	Logistic Regression	Shannon Entropy	16	**0.7552**	0.0056
Breast Cancer Wisconsin	SVM	Information Gain	7	**0.9620**	0.0053

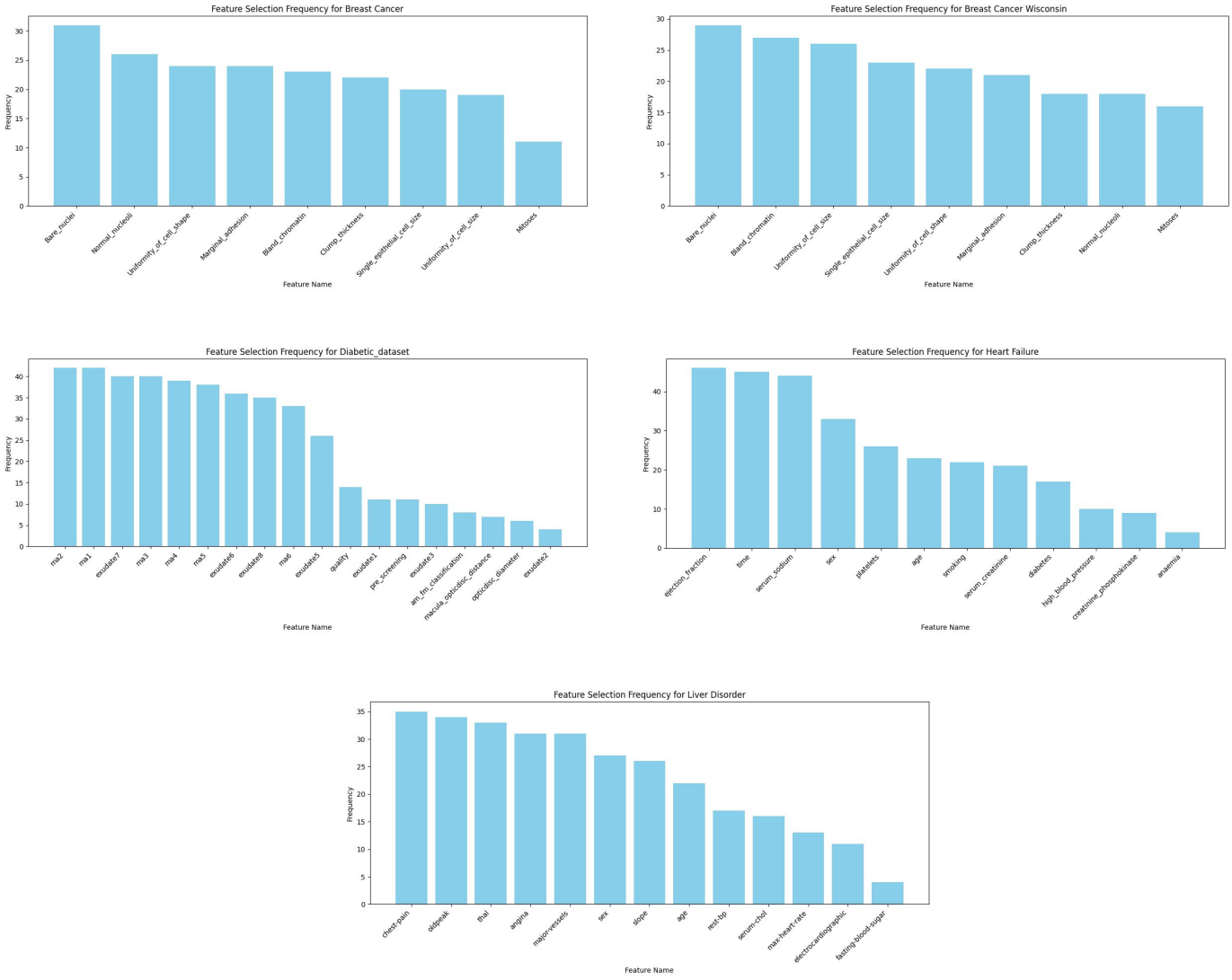


Fig. 2. Feature Selection Frequencies for Breast Cancer, Breast Cancer Wisconsin, Diabetic, Heart Failure and Liver Disorder.

TABLE VIII
CONTRIBUTION OF EACH TEAM MEMBER

Team Member	**Contribution**
Soumen Sinha	- Literature review - Formulated research question - Analyzed Diabetic and Liver Disorder datasets - Conducted T-tests - Contributed to the ablation study and final report
Denis Krylov	- Literature review - Formulated research question - Analyzed Breast Cancer and Breast Cancer Wisconsin datasets - Conducted T-tests - Contributed to the ablation study and final report
Alexandru Grigore	- Literature review - Formulated research question - Analyzed Heart faliure dataset and experimented with different scoring metrics - Conducted T-tests - Contributed to the ablation study and final report